CSE 219 — Signals and Linear Systems

Lec 1–2 · Signals: Energy, Transformations, Even–Odd, Step & Impulse, Periodicity

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Anik Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

Contents

Exam Map — What the TF Papers Ask — signal-basics questions and how much they matter
Lesson 1 — Energy and Power — E and P in CT and DT; energy vs power signals
Lesson 2 — Time Shifting, Reversal and Scaling — the x(αt+β) recipe, sketches step by step
Lesson 3 — Even and Odd Decomposition — x_e, x_o for piecewise CT and DT signals
Lesson 4 — Unit Step, Unit Impulse, Ramp and Signals Built from Them — δ = du/dt, sifting, jumps → impulses, δ models
Lesson 5 — Exponentials, Sinusoids and Periodicity — CT and DT periodicity, sums of sinusoids
Quick Notes — Signals — merged revision notes + key diagrams

> **NOTE** Slide numbers refer to Lecture-1-A-Little-Introduction-to-Signals.pdf (53 slides, “Lec 1”) and Lecture-2-Some-Useful-Signals.pdf (51 slides, “Lec 2”). Questions are labelled by session and course: CSE 219 = this course (pages in the merged TF PDF or “22 batch p.N”); EEE 211 / EEE 301 = the EEE department's Continuous Signals and Linear Systems papers, same content (pages “EEE p.N”). Ramp functions, impulse scaling and δ′(t) are textbook background (marked).

Exam Map — What the TF Papers Ask

Signal basics open almost every paper. In CSE 219 they are the first Anik-section questions (2022-23 Q5(a), Q7; 2023-24 Q1(a), Q3(a)); in the EEE papers they are Q1 or Q5 every year.

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q1(a) x(t) = 1−|t| (|t|≤1): sketch y(t) = x(−2(t−1)) + x((t+2)/2) (15) · Q3(a) x[n] = (3−4j)$e^{j13\pi n/7}$: real/imag parts periodic? N_0; phase shift as integer time shift? (15)
2023-24 (EEE 211)	Q1(a) Piecewise x(t): sketch x(−3t+2), x'(t), x''(t); x'(t) via δ, u, r (11)
2022-23 (CSE 219)	Q5(a) f(t) = t+1 (−2..0), −t/2+1 (0..2): draw, even–odd decomposition, draw parts (15) · Q5(c) Is 2cos(16πn/63 + π/3) periodic? period (5) · Q7(a)–(c) x(t) of Fig. 2: piecewise definition; sketch x(−2t+2) and x(−t/2+1) step by step (23) · Q7(d) Image-editing tasks ↔ which signal operations (7)
2022-23 (EEE 211)	Q5(c) x(t) of Fig. 5(c) via elementary signals; sketch dx/dt (10) · Q6(a) Are lim 1/(επ cosh(t/ε)) and lim sin(εt)/(πt) models of δ(t)? (20)
2021-22 (EEE 211)	Q1(a) √($e^{j5\pi t}$) + sin²(3t−π/4): periodic? period (10) · Q1(b) e^{-3t} sin(2πt/3 − 4) u(t): power or energy signal? (10) · Q1(c) x = δ(t−1)+u(t)u(t−3)+u(cos t): y = ∫x; sketch y(t/2) (15)
2020-21 (EEE 211)	Q5(a) x_1, x_2 via r, u; draw x_1(t+2)r(t−2), x_2(t−2)u(−2t); x_3 via x_1 (25) · Q5(b) Power or energy: u(t−6)−u(−t+8); 2r(t−4)u(t) (10)
2019-20 (EEE 211)	Q5(a) Single expression for x(t) of Fig. 5(a); time-compress by 3 (15)
2019-20 (CSE 311)	Q2(b) Prove u(t) from δ(t) and δ(t) from u(t) (11)
2018-19 (EEE 211)	Q6(a) Power/energy/neither: $e^{j(2\pi t+\pi/4)}$ + $e^{-2\|t\|}$; [u(cos t) − δ(t−kπ)]$^{0.5}$ (15) · Q6(b) v(t) of Fig. 6(b): steps & ramps; derivative; even and odd parts of the derivative (20)
2017-18 (EEE 211)	Q5(a) [2cos(0.6√π t+π/6) + sin(1.2√π t−π/3)]² periodic? period (15) · Q5(b) Power/energy: x(4−t/2)[δ(t+½)−δ(t−3/2)]; $\Sigma_{k=-1}^{1}$ u(2x(t)−k) (20)
2016-17 (EEE 211)	Q5(a) Sketch x(t) = (t+1)[u(t+2) − u(t)] (10) · Q5(b) y(t) = x(−2t+6) is given: sketch x(t) (15) · Q5(c) cos(100πt+π/3) + sin(40πt+2π/3) periodic? period (10)
2015-16 (EEE 301)	Q1(a) Periodic? 3cos(4t+π/3); cos²(2t−π/3); $\Sigma_{n=-7}^{7}$ $e^{-(2t-n)}$; $e^{j(\pi t-1)}$ (20) · Q1(b) Power/energy: e^{-t}cos3t u; u−2u(t−2)+u(t−5); te^{-t}u − δ(t−2k) (15) · Q2(a) Evaluate six impulse integrals/products (20)
2014-15 (EEE 301)	Q1(a) u(cos t) − δ(t−kπ): periodic? period; sketch (9) · Q1(b) $\Sigma_{k=-4}^{4}$ δ(3t−k) sin(πt/3) sgn(t): power/energy? (6) · Q1(c) x = −t+1 (−1..0), t (0..2), 2 (2..3): even/odd parts and their averages (20)
2013-14 (EEE 301)	Q1(a) x_1+x_2 periodic? 4cos πt − sin(3πt+π/2); 4cos2t + 4cos2πt (18) · Q1(b) Do lim (1/π)ε/(t²+ε²) and lim (1/π) sin(εt)/t represent δ(t)? (17) · Q2(a) Power/energy: u(t)+5u(t−1)−2u(t−2); 2 sin(3πt/8) cos(2πt/3) (18)
2012-13 (EEE 301)	Q1(a) α δ(sin t) + (1−α)δ(t−kπ), k odd: periodic? sketch for α = 0.5 (12) · Q1(b) $\Sigma_{k=-4}^{4}$ δ(3t−k) sin²(πt/3): power/energy? (12)
2012-13 (CSE 311)	Q5(a) Define unit impulse; derive the sampling (sifting) property (8)
2011-12 (EEE 301)	Q6(a) $e^{j7\pi t/6}$ + $e^{j5\pi t/6}$: periodic? period (7) · Q6(b) x = 1 (t<0), e^{3t} (t≥0): energy/power/neither (7) · Q6(c) Properties of δ(t); ∫e^{-5t+1}δ'(t−5)dt (8) · Q6(d) Sketch y = x(−2t−2) for x of Fig. 6(d); analytic expression (13)
2010-11 (EEE 301)	Q5(a)(i) cos(πt/3) + sin(πt/4): periodic? period (6) · Q5(a)(ii) Odd and even parts of A $e^{-\alpha t}$u(t) (6) · Q5(b)(i) Sketch x = u(t+2)−2u(t+1)+2u(t)−u(t−2)−2u(t−3)+2u(t−4) and y = ∫x (8) · Q5(b)(ii) Sketch x(t−2), x(t+3), x(−3t−2), x(2t/3+½) with expressions (15) · Q6(a) Power/energy: Ae^{-t}u(t); $Ae^{-a\|t\|}$ (10)

EXAM FOCUS Most repeated, in order: (1) sketch x(αt+β) or a sum of such signals — CSE 219 2022-23 Q7 (23 marks), 2023-24 Q1(a) (15), EEE every year · (2) write a piecewise signal with u(t), r(t) and sketch its derivative (impulses at jumps) · (3) periodicity of CT/DT sinusoids (2022-23 Q5(c), 2023-24 Q3(a)) · (4) even–odd parts (2022-23 Q5(a), 15 marks) · (5) energy or power signal. Learn the x(αt+β) recipe (Lesson 2) first: it is worth about 25 marks per CSE 219 paper.

Lesson 1 — Energy and Power

Slides 2–16 (Lec 1)

1. Signals (sl. 2–5)

A signal is a pattern of change that carries information, as a function of time (voice, voltage) or space (image brightness).

- **Continuous-time (CT)** signal x(t): defined at every real t (microphone voltage, temperature).
- **Discrete-time (DT)** signal x[n]: defined only at integer n (audio samples, a daily log). x[0.5] and x[π] do not exist.
- Convention: round brackets ( ) for CT, square brackets [ ] for DT.

2. Energy and Power (sl. 6–7)

Square the values and add them up. |·| makes the formulas work for complex signals too.

	Continuous time	Discrete time
Energy in an interval	$E = \int_{t1}^{t2} \lvert x(t)\rvert^2\, dt$	$E = \sum_{n=n1}^{n2} \lvert x[n]\rvert^2$
Average power in the interval	$P = E / (t2 - t1)$	$P = E / (n2 - n1 + 1)$
Total energy	$E\infty = \int_{-\infty}^{\infty} \lvert x(t)\rvert^2\, dt$	$E\infty = \sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2$
Total average power	$P\infty = \lim_{T\to\infty} (1/2T) \int_{-T}^{T} \lvert x(t)\rvert^2\, dt$	$P\infty = \lim_{N\to\infty} (1/(2N+1)) \sum_{-N}^{N} \lvert x[n]\rvert^2$

MEMORIZE **Energy signal**: $0 < E\infty < \infty$ (then $P\infty = 0$). **Power signal**: $0 < P\infty < \infty$ (then $E\infty = \infty$). **Neither**: $E\infty = \infty$ and $P\infty = \infty$ (e.g. a ramp, e^{2t}) or the integral does not exist. A signal can never be both.

• Periodic signal (period T0): $E\infty = \infty$, $P\infty = (1/T0) \int_{T0} \lvert x\rvert^2\, dt$ — one period is enough.

• Finite-duration, bounded signal: always an energy signal.

• A sinusoid $A\cos(\omega 0t + \varphi)$: $P = A^2/2$. A complex exponential $A e^{j\omega 0t}$: $P = \lvert A\rvert^2$.

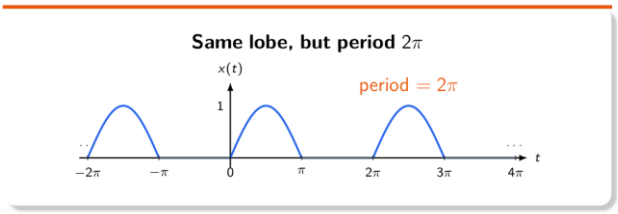
3. Worked Examples from the Slides (sl. 8–16)

Half-sine x(t) = sin t on 0 ≤ t ≤ π, 0 elsewhere (sl. 8–9):

```
E = ∫₀^π sin²t dt = ∫₀^π (1 − cos 2t)/2 dt = π/2
P[0,π] = E/(π − 0) = 1/2
```

Discrete x[0..3] = {1, −2, 2, −1} (sl. 10–11): E = 1 + 4 + 4 + 1 = 10, P[0,3] = 10/4 = 2.5.

Periodic half-sine (sl. 12–14): repeating the lobe with period π gives $E\infty = \pi/2 + \pi/2 + \ldots = \infty$ but $P = (1/\pi)(\pi/2) = 1/2$. If the lobe rests for half the period (period 2π), $P = (1/2\pi)(\pi/2) = 1/4$.



Answer:

$$E = \infty, \qquad P = \frac{1}{2\pi}\int_0^{\pi} \sin^2 t\, dt = \frac{1}{4}.$$

Figure 1.1 — Half-sine repeating with period 2π: infinite energy, power 1/4 (Lec 1, sl. 14)

An infinite-range signal with finite energy (sl. 15–16): x(t) = 1 for |t| ≤ 1, 1/|t| for |t| > 1.

```
E = ∫_{-1}^{1} 1 dt + 2 ∫₁^∞ t^{-2} dt = 2 + 2[-1/t]₁^∞ = 2 + 2 = 4
P∞ = 0     (finite energy spread over infinite time)
```

TRAP Infinite duration does not mean infinite energy: 1/|t| tails have finite energy because $\int t^{-2}$ converges. But 1/√|t| tails would give $\int t^{-1} = \infty$. Always integrate |x|², not x.

Exam Approach

1. Sketch |x|² and decide what happens as t → ±∞: decays fast enough → try energy; settles to a nonzero level or is periodic → power; grows → neither.
2. Energy: integrate |x|² over the support (split piecewise signals).
3. Power: for periodic signals average one period; for signals that settle to constants c_- (t → −∞) and c_+ (t → +∞), P∞ = $(c_-^2 + c_+^2)/2$.
4. State the class in words: “E finite ⇒ energy signal, P = 0”.

Practice — Lesson 1

(a) Actual — EEE 211 2021-22 Q1(b). (b) Actual — EEE 211 2020-21 Q5(b). (c) Actual — EEE 211 2018-19 Q6(a)(i). (d) Practice set 1.1. (e) Older EEE papers, quick answers.

(a) Determine whether $e^{-3t}\sin(2\pi t/3 - 4)\,u(t)$ is a power or an energy signal. (10)

(b) Power or energy: (i) u(t − 6) − u(−t + 8); (ii) 2r(t − 4)u(t). (10)

(c) Is $x_1(t) = \exp[j(2\pi t + \pi/4)] + \exp(-2|t|)$ a power or an energy signal? (part of 15)

(d) x(t) = t + 1 (−1 ≤ t < 0), 2 − t (0 ≤ t ≤ 2), 0 otherwise. Find its energy and its average power over [−1, 2].

(e) Classify: (i) $e^{-t}\cos 3t\,u(t)$; (ii) u(t) − 2u(t−2) + u(t−5); (iii) u(t) + 5u(t−1) − 2u(t−2); (iv) $2\sin(3\pi t/8)\cos(2\pi t/3)$; (v) 1 for t < 0, e^{3t} for t ≥ 0; (vi) $A\,e^{-t}u(t)$ and $A\,e^{-a|t|}$ (a > 0).

✓ Solution

(a) $|x|^2 \le e^{-6t}$ for t ≥ 0, so $E \le \int_0^\infty e^{-6t}\,dt = 1/6 < \infty$ → energy signal (P∞ = 0). Exact value (write $\sin^2 = (1 - \cos 2\theta)/2$ with ω = 2π/3):

```
E = ∫₀^∞ e^{-6t} sin²(ωt − 4) dt = 1/12 − (1/2) Re{ e^{-j8} / (6 − j2ω) }
  = 1/12 − (6 cos 8 + 2ω sin 8) / (2(36 + 4ω²))
  = 0.08333 − (−0.873 + 4.144)/107.09 = 0.0528
```

(b) (i) u(t − 6) = 1 for t > 6; u(8 − t) = 1 for t < 8. So x = −1 (t < 6), 0 (6 < t < 8), +1 (t > 8). E∞ = ∞, P∞ = [(−1)² + 1²]/2 = 1 → **power signal**, P = 1. (ii) 2r(t − 4)u(t) = 2(t − 4) for t ≥ 4: it grows without bound, (1/2T)∫(2(t−4))²dt → ∞ → **neither**.

(c) $|x_1|^2 = 1 + e^{-4|t|} + 2e^{-2|t|}\cos(2\pi t + \pi/4)$. The last two terms have finite area, so their contribution to $(1/2T)\int$ vanishes: $P_\infty = 1$, $E_\infty = \infty \rightarrow$ **power signal**, $P = 1$ (the power of the exponential).

(d) Integrate each piece:

```
E = ∫_{-1}^{0} (t+1)² dt + ∫₀² (2-t)² dt = 1/3 + 8/3 = 3
P[-1,2] = E / 3 = 1
```

(e) (i) $E = \int_0^\infty e^{-2t}(1 + \cos 6t)/2\, dt = ½(½ + 2/40) = 0.275 \rightarrow$ energy. (ii) 1 on (0,2), −1 on (2,5): $E = 2 + 3 = 5 \rightarrow$ energy. (iii) value 6 on (1,2), 4 for $t > 2$: $P_\infty = 4^2/2 = 8 \rightarrow$ power. (iv) $= \sin(25\pi t/24) - \sin(7\pi t/24)$: two sinusoids of amplitude 1 $\rightarrow P = ½ + ½ = 1 \rightarrow$ power. (v) grows as $e^{3t} \rightarrow$ neither. (vi) $E = A^2/2$ and $E = 2 \cdot A^2/(2a) = A^2/a \rightarrow$ both energy signals.

> **NOTE** EEE 301 2015-16 Q1(b)(iii) and EEE 211 2018-19 Q6(a)(ii) contain impulses $\delta(t - 2k)$ inside the signal. $|\delta|^2$ is not defined, so the energy integral does not exist: classify as “neither” and say why.

Lesson 2 — Time Shifting, Reversal and Scaling

Slides 17–39 (Lec 1)

1. The Three Operations (sl. 18–25)

Operation	Rule (t0 > 0, α > 1)	Effect on a feature at t = a
Delay	x(t − t0)	moves right: a → a + t0
Advance	x(t + t0)	moves left: a → a − t0
Reversal	x(−t)	mirror about t = 0: a → −a
Compression	x(αt)	a → a/α (narrower)
Stretching	x(t/α)	a → αa (wider)
Amplitude scaling	A·x(t)	height × A, time unchanged

DT shifts work the same way: x[n − 4] moves every sample 4 places right (sl. 21). Time scaling changes only the horizontal axis, not the height (sl. 25).

2. Order Matters (sl. 26–27)

```
x(t) --shift left 3--> x(t+3) --compress by 2--> x(2t+3)      ✓
x(t) --compress by 2--> x(2t) --shift left 3--> x(2(t+3)) = x(2t+6)   ✗ (different!)
```

Each operation replaces t by something in the **whole** current expression. After compressing, a later shift of 3 becomes a shift of the argument by 6.

3. The Recipe for x(αt + β) (sl. 28–31)

$$\mathbf{x(t) \rightarrow g_1(t) = x(t + \beta) \rightarrow g_2(t) = g_1(-t) \text{ if } \alpha < 0 \rightarrow g_3(t) = g_2(|\alpha| t) = x(\alpha t + \beta)}$$

- Step 1 — shift by β (β > 0: left; β < 0: right).
- Step 2 — reverse only if α < 0.
- Step 3 — scale by |α| (|α| > 1 compress, |α| < 1 stretch).
- Write the intermediate signal after every step (sl. 30).

MEMORIZE Fast check (no sketch needed): a feature of x at t = a appears in x(αt + β) at **t = (a − β)/α**. Use it to find the new break points and the new support, then join them with the same shapes.

4. Worked Example: x(−3t + 2) (sl. 31–35)

x(t) = 1 on 0 ≤ t < 1, 2 − t on 1 ≤ t ≤ 2, 0 otherwise.

```
Step 1  x(t+2)   = 1 on [-2,-1),  -t on [-1,0]
Step 2  x(-t+2)  = t on [0,1),    1 on [1,2]
Step 3  x(-3t+2) = 3t on [0,1/3), 1 on [1/3,2/3]
Check with t = (a - β)/α: a = 0 → t = 2/3; a = 1 → t = 1/3; a = 2 → t = 0  ✓
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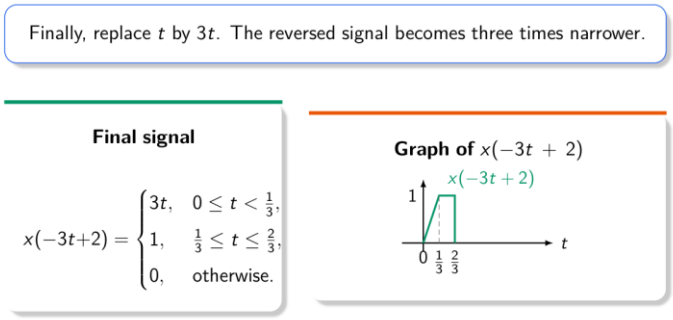


Figure 2.1 — The final compressed, reversed signal x(−3t + 2) (Lec 1, sl. 35)

5. Two Practice Problems from the Slides (sl. 36–39)

Practice 1: x(3(2 − t/2)) = x(6 − 3t/2): shift left by 6 → x(t + 6); reverse → x(−t + 6); compress by 3/2 → x(−3t/2 + 6).

Practice 2: start with x(t) and stretch by 2, shift left by 3, compress by 4, reverse:

```
y1 = x(t/2);  y2 = y1(t+3) = x((t+3)/2);  y3 = y2(4t) = x(2t + 3/2);  y4 = y3(-t) = x(-2t + 3/2)
```

TRAP • A shift applied after scaling acts on t, not on αt: from x(2t), "shift left by 3" gives x(2(t+3)). • Reversal and scaling commute, but neither commutes with a shift. • Keep jumps (vertical edges) as jumps — they only move, they never become slopes.

Exam Approach

1. Write x(t) piecewise first (the papers give 3 marks just for this).
2. Rewrite the argument as αt + β (expand brackets: −2(t − 1) = −2t + 2).
3. Apply the recipe: shift, reverse if α < 0, scale; name each intermediate signal and the property used.
4. Map every break point with t = (a − β)/α; check one interior value by direct substitution.
5. For sums such as x(·) + x(·), sketch each term separately, then add point by point.

Practice — Lesson 2

(a) Actual — CSE 219 2022-23 Q7(a)–(c). (b) Actual — CSE 219 2023-24 Q1(a). (c) Actual — CSE 219 2022-23 Q7(d). (d) Actual — EEE 301 2010-11 Q5(b)(ii). (e) Adapted — EEE 211 2016-17 Q5(b).

(a) x(t) of Figure 2 is 0 for t < −2 and t > 2: it rises linearly from −1 at t = −2 to 0 at t = −1, jumps to 1 and stays at 1 up to t = 0, then falls linearly to 0 at t = 2. (i) Write x(t) piecewise. (ii) Determine and draw x(−2t + 2), naming the properties used. (iii) Determine and draw x(−t/2 + 1). (3 + 10 + 10)

(b) x(t) = 1 − |t| for |t| ≤ 1, 0 otherwise. Sketch y(t) = x(−2(t − 1)) + x((t + 2)/2), showing the effect of each operation on the two components. (15)

(c) An image is a discrete signal. Which signal property/operation is useful for: compression, blurring, sharpening, flipping, zooming, contrast, edge detection? (7)

(d) x(t) = −t + 1 (−1 ≤ t < 0), t (0 ≤ t < 2), 2 (2 ≤ t < 3), 0 otherwise. Find x(t − 2), x(t + 3), x(−3t − 2) and x(2t/3 + 1/2) analytically.

(e) If y(t) = x(−2t + 6) is known, how do you recover x(t)?

✓ Solution

(a) (i) x(t) = t + 1 for −2 ≤ t < −1; 1 for −1 ≤ t ≤ 0; 1 − t/2 for 0 < t ≤ 2; 0 otherwise.

```
(ii) x(−2t+2):  α = −2, β = 2
Step 1 (time shift, left by 2):  g1 = x(t+2) = t+3 on [−4,−3),  1 on [−3,−2],  −t/2 on (−2,0]
Step 2 (time reversal):          g2 = x(−t+2) = t/2 on [0,2),  1 on [2,3],  3−t on (3,4]
Step 3 (time compression by 2):  x(−2t+2) = t on [0,1),  1 on [1,1.5],  3−2t on (1.5,2]
Check: t = 2 → x(−2) = −1 and 3 − 4 = −1 ✓;  t = 1 → x(0) = 1 ✓

(iii) x(−t/2+1):  α = −1/2, β = 1
Step 1 (shift left 1):     x(t+1) = t+2 on [−3,−2),  1 on [−2,−1],  (1−t)/2 on (−1,1]
Step 2 (reversal):         x(−t+1) = (1+t)/2 on [−1,1),  1 on [1,2],  2−t on (2,3]
Step 3 (stretch by 2):     x(−t/2+1) = 1/2 + t/4 on [−2,2),  1 on [2,4],  2 − t/2 on (4,6]
Check: t = 6 → x(−2) = −1 and 2 − 3 = −1 ✓;  t = −2 → x(2) = 0 ✓
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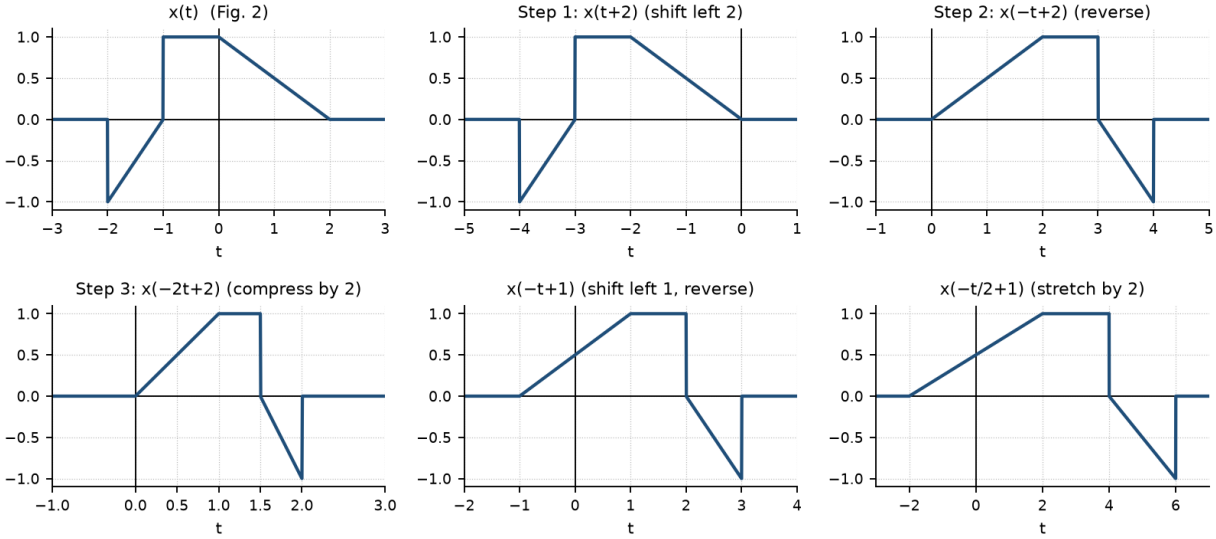


Figure 2.2 — CSE 219 2022-23 Q7: x(t), the three steps to x(−2t+2), and x(−t/2+1)

(b) First component x(−2(t − 1)) = x(−2t + 2): shift left 2 → triangle centred at −2; reverse → centred at +2; compress by 2 → centred at t = 1, half-width 1/2 (feature at a = 0 → t = (0 − 2)/(−2) = 1; a = ±1 → t = ½, 3/2). Reversal changes nothing visible because the triangle is even. Second component x((t + 2)/2) = x(t/2 + 1): shift left 1 → centred at −1; stretch by 2 → centred at t = −2, half-width 2 (a = ±1 → t = 0, −4). The two triangles do not overlap, so:

```
y(t) = (t+4)/2      −4 ≤ t ≤ −2
       −t/2         −2 < t ≤ 0
       2t − 1        1/2 ≤ t ≤ 1
       3 − 2t        1 < t ≤ 3/2
       0             otherwise        (peaks y(−2) = y(1) = 1)
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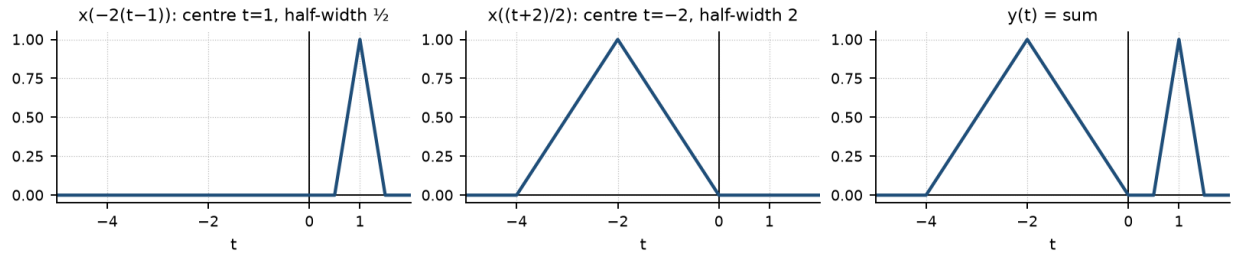


Figure 2.3 — CSE 219 2023-24 Q1(a): the two transformed triangles and their sum

(c) Compression — frequency content (Fourier/DCT): keep large low-frequency coefficients, drop small high-frequency ones. Blurring — convolution with an averaging (low-pass) kernel. Sharpening — high-pass filtering / adding back a differentiated (difference) image. Flipping — time (index) reversal x[−n] (x[−m, n] for a horizontal flip). Zooming — time (index) scaling: up-/down-sampling with interpolation. Contrast — amplitude scaling (and shifting) of pixel values; the spread (power/variance) of the values measures it. Edge detection — first difference x[n] − x[n−1] (derivative), i.e. high frequencies.

(d) Map each interval endpoint with t = (a − β)/α and keep the formula of each piece:

```
x(t−2)   : shift right 2:  −t+3 on [1,2),  t−2 on [2,4),  2 on [4,5)
x(t+3)   : shift left 3 :  −t−2 on [−4,−3), t+3 on [−3,−1), 2 on [−1,0)
x(−3t−2) : a = −3t−2, t = −(a+2)/3:
           a ∈ [−1,0) → t ∈ (−2/3,−1/3]: x = −a+1 = 3t+3
           a ∈ [0,2)  → t ∈ (−4/3,−2/3]: x = a = −3t−2
           a ∈ [2,3)  → t ∈ (−5/3,−4/3]: x = 2
x(2t/3+1/2): a = 2t/3+1/2, t = 3(a − 1/2)/2:
           a ∈ [−1,0) → t ∈ [−9/4,−3/4): x = −a+1 = 1/2 − 2t/3
           a ∈ [0,2)  → t ∈ [−3/4, 9/4): x = 2t/3 + 1/2
           a ∈ [2,3)  → t ∈ [9/4, 15/4): x = 2
```

(e) Undo the operations in reverse order: y(t) = x(−2t + 6) ⇒ x(t) = y((6 − t)/2) = y(−t/2 + 3). Recipe on y: shift left by 3, reverse, stretch by 2. Every feature of y at t = b goes to t = 6 − 2b.

Lesson 3 — Even and Odd Decomposition

Slides 40–52 (Lec 1)

1. Definitions (sl. 41–44)

- Even: x(−t) = x(t) (mirror about the vertical axis), e.g. cos t.
- Odd: x(−t) = −x(t) (point symmetry about the origin), e.g. sin t. An odd signal has x(0) = 0.

$$x_e(t) = \tfrac{1}{2}[x(t) + x(-t)] \qquad x_o(t) = \tfrac{1}{2}[x(t) - x(-t)]$$
$$x(t) = x_e(t) + x_o(t)$$

Checks (sl. 44): x_e(−t) = ½[x(−t) + x(t)] = x_e(t); x_o(−t) = ½[x(−t) − x(t)] = −x_o(t); adding them gives x(t). The same formulas hold in DT with n in place of t.

2. Worked Examples (sl. 45–52)

x(t) = t + 1 on −1 ≤ t ≤ 1: x(−t) = 1 − t, so x_e = ½[(t+1) + (1−t)] = 1 and x_o = ½[(t+1) − (1−t)] = t, both on |t| ≤ 1.

DT step x[n] = u[n]: x_e[n] = ½ for n ≠ 0 and 1 at n = 0; x_o[n] = −½ (n < 0), 0 (n = 0), ½ (n > 0).

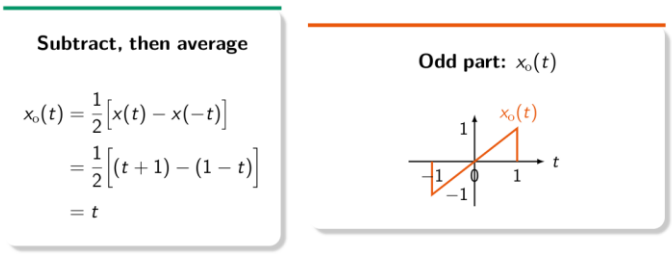


Figure 3.1 — Odd part of x(t) = t + 1 on [−1, 1] is x_o(t) = t (Lec 1, sl. 48)

MEMORIZE Useful facts: even × even = even, odd × odd = even, even × odd = odd. $\int_{-a}^{a}$ odd = 0. The area of x_e equals the area of x (the odd part adds zero area). x_e(0) = x(0), x_o(0) = 0.

TRAP Work interval by interval: x(−t) has its pieces on the mirrored intervals, so on each sub-interval you must combine the correct piece of x(t) with the correct piece of x(−t) (one of them may be 0). Do not just replace t by −t inside one formula.

Exam Approach

1. Write x(t) piecewise and list all break points ±a.
2. Write x(−t) piecewise (mirror the intervals, replace t by −t in each formula).
3. Cut the axis at all break points of both and compute ½(x ± x(−t)) on each piece.
4. Sketch: x_e must be mirror-symmetric, x_o must be point-symmetric — use this as a check.

Practice — Lesson 3

(a) Actual — CSE 219 2022-23 Q5(a). (b) Actual — EEE 301 2014-15 Q1(c). (c) Actual — EEE 301 2010-11 Q5(a)(ii).

(a) f(t) = t + 1 (−2 ≤ t ≤ 0), −t/2 + 1 (0 < t ≤ 2), 0 elsewhere. (i) Draw f(t). (ii) Find its even–odd decomposition using the definitions. (iii) Draw both parts. (3 + 6 + 6)

(b) x(t) = −t + 1 (−1 ≤ t < 0), t (0 ≤ t < 2), 2 (2 ≤ t < 3), 0 otherwise. Find the odd and even parts and their average values. (20)

(c) Find and sketch the odd and even parts of x(t) = A $e^{-\alpha t}$, t > 0, and 0 otherwise.

✓ Solution

(a) f(−t) = −t + 1 for 0 ≤ t ≤ 2 and t/2 + 1 for −2 ≤ t < 0.

```
For 0 < t ≤ 2:  f_e = ½[(-t/2 + 1) + (-t + 1)] = 1 − 3t/4     f_o = ½[(-t/2 + 1) − (-t + 1)] =
t/4
For −2 ≤ t < 0: f_e = ½[(t + 1) + (t/2 + 1)] = 1 + 3t/4     f_o = ½[(t + 1) − (t/2 + 1)] = t/4

f_e(t) = 1 − 3|t|/4,  |t| ≤ 2  (0 elsewhere)        f_o(t) = t/4,  |t| ≤ 2  (0 elsewhere)
Check: f_e + f_o at t = 1 → 1/4 + 1/4 = 1/2 = −1/2 + 1 ✓;  at t = −1 → 1/4 − 1/4 = 0 = −1 + 1
✓
```

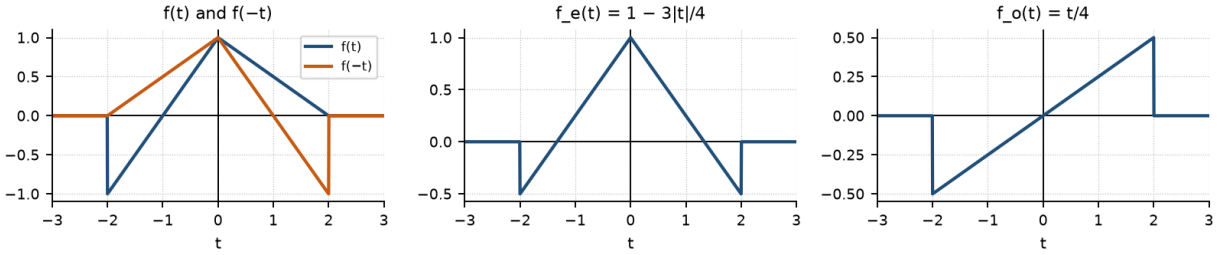


Figure 3.2 — CSE 219 2022-23 Q5(a): f(t), its even part (a 'tent') and its odd part (a line)

(b) x(−t) = t + 1 on (0, 1], −t on (−2, 0], 2 on (−3, −2]. Combining interval by interval:

```
interval     x(t)    x(-t)    x_e = ½(x + x(-t))    x_o = ½(x − x(-t))
(-3,-2)       0       2          1                    -1
(-2,-1)       0      -t         -t/2                   t/2
(-1, 0)     -t+1     -t         -t + 1/2               1/2
( 0, 1)       t      t+1         t + 1/2              -1/2
( 1, 2)       t       0          t/2                   t/2
( 2, 3)       2       0          1                     1
```

Average values over the support (−3, 3): the odd part averages to **0** (odd function on a symmetric interval). The even part has the same area as x: ∫x = 1.5 + 2 + 2 = 5.5, so its average is 5.5/6 ≈ **0.917**. (Over all time both averages are 0, because the signal has finite duration.)

(c) x_e(t) = (A/2) $e^{-\alpha|t|}$ (all t); x_o(t) = (A/2) $e^{-\alpha t}$ for t > 0 and −(A/2) $e^{\alpha t}$ for t < 0, i.e. (A/2) sgn(t) $e^{-\alpha|t|}$. At t = 0: x_e = A/2 (taking x(0) = A/2), x_o = 0.

Lesson 4 — Unit Step, Unit Impulse, Ramp and Signals Built from Them

Slides 2–29 (Lec 2) · ramp, impulse scaling and δ'(t) are textbook background

1. Discrete-Time Step and Impulse (sl. 3–15)

u[n] = 1 (n ≥ 0), 0 (n < 0) δ[n] = 1 (n = 0), 0 (n ≠ 0)

- R1 — first difference: δ[n] = u[n] − u[n − 1] (sl. 11).
- R2 — running sum: u[n] = $\Sigma_{m=-\infty}^{n}$ δ[m] — the spike at m = 0 is included only when n ≥ 0 (sl. 12–13).
- R3 — sum of delayed impulses: u[n] = $\Sigma_{k=0}^{\infty}$ δ[n − k] (sl. 14–15).
- Values: u[1] = 1, u[−3] = 0, u[0] = 1; δ[0] = 1, δ[4] = δ[−2] = 0 (sl. 7–8). u[n − 2] starts at n = 2, u[n + 3] at n = −3 (sl. 9–10).

2. Continuous-Time Step and Impulse (sl. 16–27)

u(t) = 0 for t < 0, 1 for t > 0 (the value at t = 0 does not matter). The impulse should be the derivative of the step, but u(t) has no ordinary derivative at the jump. Fix (sl. 20–25): replace the jump by a short ramp u_Δ(t) rising from 0 to 1 over [0, Δ]. Its derivative δ_Δ(t) is a pulse of height 1/Δ and width Δ — area 1 for every Δ. As Δ → 0 the pulse becomes the ideal impulse:

δ(t) = du(t)/dt u(t) = $\int_{-\infty}^{t}$ δ(τ) dτ $\int$ δ(t) dt = 1

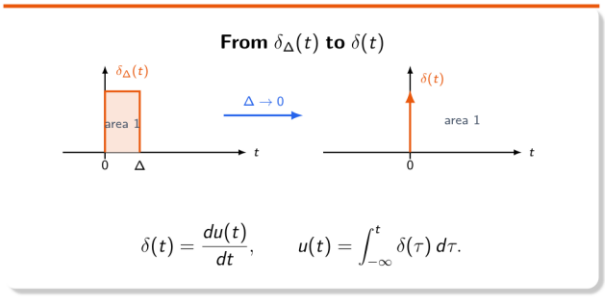


Figure 4.1 — δ(t) is the limit of unit-area pulses δ_Δ(t) as Δ → 0 (Lec 2, sl. 25)

Arrow notation (sl. 26): the number beside the arrow is the **area (strength)**, not the height; 2δ(t − 1) is an arrow of strength 2 at t = 1, −3δ(t − 3) points down. d/dt[2u(t)] = 2δ(t) (sl. 27).

3. Signals Made of Steps; Derivatives at Jumps (sl. 28–29)

Read the jumps from left to right: x(t) jumps +2 at t = 1, −3 at t = 2 and +2 at t = 4, so

```
x(t) = 2u(t−1) − 3u(t−2) + 2u(t−4)
dx/dt = 2δ(t−1) − 3δ(t−2) + 2δ(t−4)     (each jump → an impulse equal to the jump height)
```

$$\frac{dx(t)}{dt} = 2\delta(t-1) - 3\delta(t-2) + 2\delta(t-4)$$

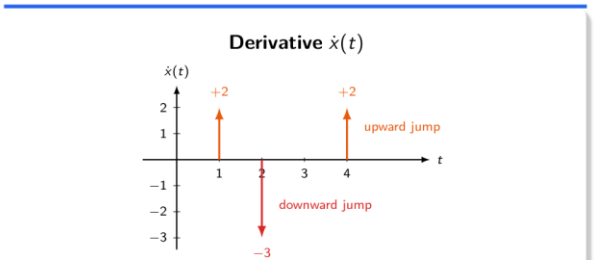


Figure 4.2 — Derivative of a jump signal: upward jumps give positive impulses (Lec 2, sl. 29)

4. Ramp and More Impulse Rules (textbook)

Rule	Statement
Ramp	$r(t) = t\,u(t) = \int_{-\infty}^{t} u(\tau)d\tau$; $dr/dt = u(t)$
Sampling (product)	$x(t)\,\delta(t-a) = x(a)\,\delta(t-a)$
Sifting	$\int_L^R x(t)\,\delta(t-a)\,dt = x(a)$ if $L < a < R$, 0 if a is outside (Lec 6, sl. 45–51)
Scaling	$\delta(at+b) = (1/\lvert a\rvert)\,\delta(t+b/a)$; $\delta(-t) = \delta(t)$ (even)
Composite	$\delta(f(t)) = \Sigma\,\delta(t-t_i)/\lvert f'(t_i)\rvert$ over the simple zeros t_i of f
Doublet	$\int x(t)\,\delta'(t-a)\,dt = -x'(a)$; δ' is odd: $\delta'(-t) = -\delta'(t)$
Building blocks	slope change of m at $t = a \rightarrow m\cdot r(t-a)$; jump of J at $t = a \rightarrow J\cdot u(t-a)$

Model of δ(t) (needed in EEE 2022-23 Q6(a), 2013-14 Q1(b)): a family $p_\varepsilon(t)$ represents $\delta(t)$ as $\varepsilon \rightarrow 0$ if (1) its area is 1 for every ε, (2) $p_\varepsilon(t) \rightarrow 0$ for every $t \neq 0$, and (3) it becomes infinitely tall at $t = 0$ (all the area concentrates at the origin).

MEMORIZE Piecewise linear signal → elementary signals: scan left to right; at each break point add (new slope − old slope)·r(t − a) and (jump)·u(t − a). Derivative: replace r by u and u by δ.

TRAP • u(3t) = u(t) and u(−2t) = u(−t): scaling a step by a positive number changes nothing. • δ(3t − k) = (1/3)δ(t − k/3), not δ(t − k/3). • Impulses at a jump are part of the derivative — the answer “slope only” loses most of the marks.

Exam Approach

1. Tabulate break points, slopes and jumps from the figure.
2. Write $x(t) = \Sigma\,(\text{slope changes})\cdot r(t-a) + \Sigma\,(\text{jumps})\cdot u(t-a)$; check two values.
3. Derivative: piecewise slopes (rectangles) + impulses of strength = jump; second derivative: impulses at slope changes.
4. Impulse integrals: find where the argument of δ is zero, scale by $1/\lvert a\rvert$, check the point is inside the limits, then sift.

Practice — Lesson 4

(a) Actual — EEE 211 2023-24 Q1(a). (b) Actual — EEE 211 2022-23 Q5(c). (c) Actual — EEE 211 2020-21 Q5(a). (d) Actual — EEE 211 2019-20 Q5(a).

(a) x(t) is a triangle from (−2, 0) up to (−1, 1) and down to (0, 0), then a ramp from (0, 0) to (2, 2), flat at 2 until t = 3, and a line down to 0 at t = 4. (i) Draw x(−3t + 2), x′(t), x″(t). (ii) Express x′(t) using δ(t), u(t), r(t). (11)

(b) x(t) = 5t on (0, 1), −6 on (1, 2), 5(t − 2) on (2, 3), 0 elsewhere. Write x(t) as a combination of elementary signals and sketch dx/dt. (10)

(c) $x_1(t)$: ramp from 0 to 2 on [0, 2], then 2 until t = 6. $x_2(t)$: ramp from 0 at t = −4 to 2 at t = −2, 2 until t = 2, 3 on (2, 4). $x_3(t)$: rises from 0 at t = −3 to 4 at t = −2, stays 4 until t = 0, then 1 until t = 2, then falls to 0 at t = 3. (i) Express x_1, x_2 with r and u. (ii) Draw $x_1(t + 2)r(t − 2)$ and $x_2(t − 2)u(−2t)$. (iii) Express x_3 in terms of x_1. (10 + 10 + 5)

(d) x(t) = 2 on (−1.5, 0), $2e^{-t/2}$ on (0, 3), 0 elsewhere. Describe x(t) by a single expression valid for all t and sketch x(3t). (15)

✓ Solution

```
(a) x(t) = t+2 [−2,−1];  −t [−1,0];  t [0,2];  2 [2,3];  8−2t [3,4];  0 otherwise
(i) x(−3t+2): a feature at a goes to t = (2 − a)/3:
    a = 4,3,2,0,−1,−2  →  t = −2/3, −1/3, 0, 2/3, 1, 4/3   with values 0, 2, 2, 0, 1, 0
    x(−3t+2) = 6t+4 [−2/3,−1/3];  2 [−1/3,0];  2−3t [0,2/3];  3t−2 [2/3,1];  4−3t [1,4/3]
    x′(t) = 1 (−2,−1);  −1 (−1,0);  1 (0,2);  0 (2,3);  −2 (3,4)       (continuous x ⇒ no
impulses)
    x″(t) = δ(t+2) − 2δ(t+1) + 2δ(t) − δ(t−2) − 2δ(t−3) + 2δ(t−4)
(ii) x′(t) = u(t+2) − 2u(t+1) + 2u(t) − u(t−2) − 2u(t−3) + 2u(t−4)
     [equivalently x(t) = r(t+2) − 2r(t+1) + 2r(t) − r(t−2) − 2r(t−3) + 2r(t−4)]
```

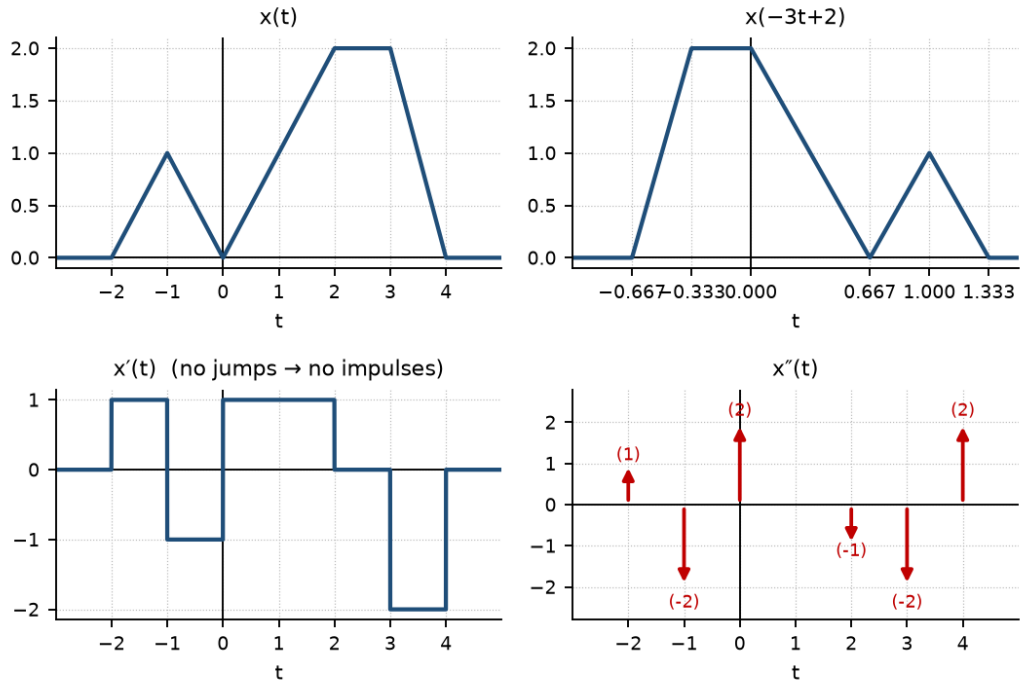


Figure 4.3 — EEE 211 2023-24 Q1(a): x(t), x(−3t+2), x'(t) and x"(t)

```
(b) Scan left to right: slope 5 starts at 0; at t = 1 slope 5 → 0 and jump 5 → −6 (−11);
    at t = 2 jump −6 → 0 (+6) and slope 0 → 5; at t = 3 slope 5 → 0 and jump 5 → 0 (−5).
    x(t) = 5r(t) − 5r(t−1) − 11u(t−1) + 6u(t−2) + 5r(t−2) − 5r(t−3) − 5u(t−3)
    dx/dt = 5[u(t) − u(t−1)] + 5[u(t−2) − u(t−3)] − 11δ(t−1) + 6δ(t−2) − 5δ(t−3)
```

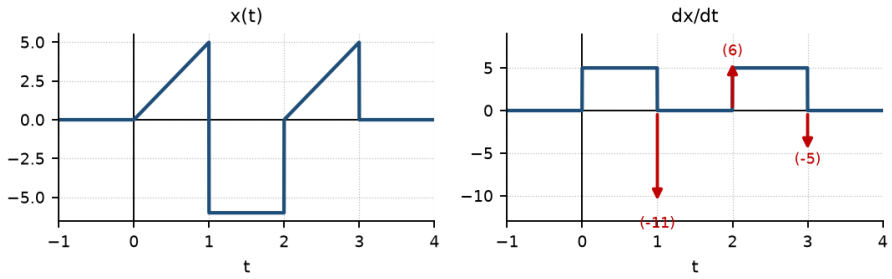


Figure 4.4 — EEE 211 2022-23 Q5(c): x(t) and dx/dt (rectangles of height 5 plus three impulses)

```
(c)(i)  x₁(t) = r(t) − r(t−2) − 2u(t−6)
        x₂(t) = r(t+4) − r(t+2) + u(t−2) − 3u(t−4)
(ii) x₁(t+2) = ramp on [−2,0], 2 on [0,4];  r(t−2) = t−2 for t ≥ 2  ⇒ product = 2(t−2) on
[2,4], 0 elsewhere
     x₂(t−2) = ramp 0→2 on [−2,0], 2 on [0,4], 3 on [4,6];  u(−2t) = u(−t) keeps t < 0  ⇒
product = t+2 on [−2,0]
(iii) Left part (ramp over 1 unit to 4, flat 2 units) = 2·x₁(2(t+3)) = 2x₁(2t+6)
      Right part (1 on [0,2], ramp down to 0 at 3) = ½·x₁(−2(t−3)) = ½x₁(6−2t)
      x₃(t) = 2x₁(2t+6) + ½x₁(6−2t)     check: t = −1 → 2·2 + 0 = 4;  t = 2.5 → 0 + ½·1 = 0.5 ✓
```

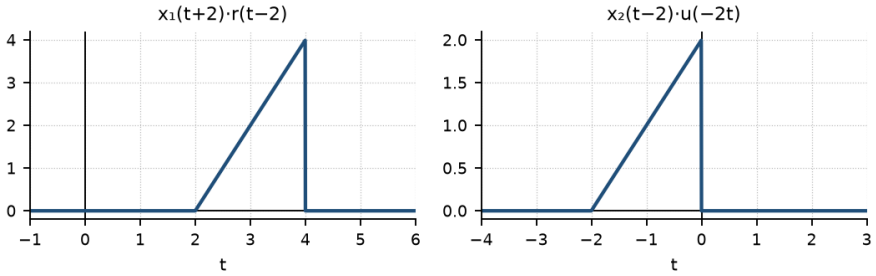


Figure 4.5 — EEE 211 2020-21 Q5(a)(ii): the two products

```
(d) x(t) = 2[u(t+1.5) − u(t)] + 2e^{−t/2}[u(t) − u(t−3)]
    x(3t) = 2[u(t+0.5) − u(t)] + 2e^{−3t/2}[u(t) − u(t−1)]     (break points −1.5, 0, 3 → −0.5,
0, 1)
```

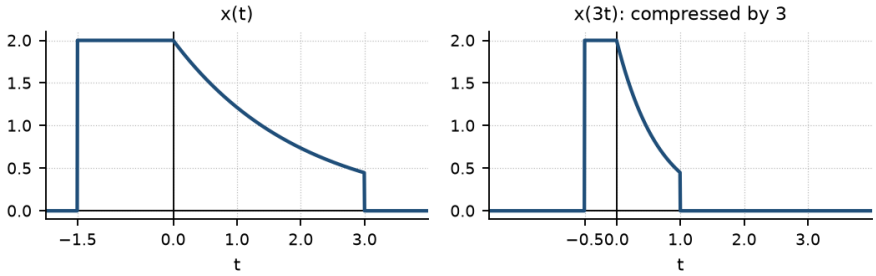


Figure 4.6 — EEE 211 2019-20 Q5(a): x(t) and x(3t)

Practice — Lesson 4 (continued)

(e) Actual — EEE 211 2022-23 Q6(a) and EEE 301 2013-14 Q1(b). (f) Actual — CSE 311 2019-20 Q2(b) and 2012-13 Q5(a). (g) Actual — EEE 211 2021-22 Q1(c). (h) Actual — EEE 301 2015-16 Q2(a), 2011-12 Q6(c), 2012-13 Q1(b).

(e) Check whether $p_1(t) = \lim_{\varepsilon\to 0} 1/(\varepsilon\pi \cosh(t/\varepsilon))$ and $p_2(t) = \lim_{\varepsilon\to 0} (1/\pi)\sin(\varepsilon t)/t$ can be used as models of $\delta(t)$. Same for $(1/\pi)\,\varepsilon/(t^2+\varepsilon^2)$. (20)

(f) Prove that a unit step can be derived from a unit impulse and vice versa. Define the unit impulse and derive its sampling property. (11; 8)

(g) $x(t) = \delta(t-1) + u(t)u(t-3) + u(\cos t)$. Compute $y(t) = \int x(\tau)d\tau$ and sketch $y(t/2)$. (15)

(h) Evaluate: $\int(t^2 + 3t + 2)\delta(-1 - t)dt$; $\int e^{t-1}\sin\{\pi(t-5)/2\}\,\delta(t + 3)dt$; $((j\omega+2)/(\omega^2+9))\delta(\omega)$; $\delta(\omega-3)/(j\omega+2)$; $\int_{-5}^{5} u_2(1 - \tau)\cos 2\pi\tau\, d\tau$ with $u_2 = d^2u/dt^2$; $\int e^{-5t+1}\delta'(t - 5)dt$. Also: is $\Sigma_{k=-4}^{4}\,\delta(3t - k)\sin^2(\pi t/3)$ a power or an energy signal?

✓ Solution

(e) Test the three conditions. p_1: substitute $u = t/\varepsilon$: $\int dt/(\varepsilon\pi\cosh(t/\varepsilon)) = (1/\pi)\int du/\cosh u = (1/\pi)\cdot\pi = 1$ for every ε. For $t \neq 0$, $\cosh(t/\varepsilon) \approx \tfrac{1}{2}e^{|t|/\varepsilon} \to \infty$ much faster than $1/\varepsilon$ grows, so $p_1 \to 0$; $p_1(0) = 1/(\varepsilon\pi) \to \infty$. **$p_1$ is a valid model of δ(t).** p_2: area $\int \sin(\varepsilon t)/(\pi t)\,dt = 1$ for every $\varepsilon > 0$, but as $\varepsilon \to 0$, $\sin(\varepsilon t)/(\pi t) \to 0$ for every t (even the peak $\varepsilon/\pi \to 0$) — the unit area spreads over a main lobe of width $2\pi/\varepsilon \to \infty$. **$p_2$ is not a model as $\varepsilon \to 0$** (it does become δ(t) as $\varepsilon \to \infty$). Lorentzian $(1/\pi)\varepsilon/(t^2 + \varepsilon^2)$: area $(1/\pi)[\tan^{-1}(t/\varepsilon)]_{-\infty}^{\infty} = 1$; $\to 0$ for $t \neq 0$; peak $1/(\pi\varepsilon) \to \infty \Rightarrow$ **valid model**.

(f) Define δ(t) as the limit of the unit-area pulse $\delta_\Delta(t)$ (height $1/\Delta$ on $[0, \Delta]$) as $\Delta \to 0$: $\delta(t) = 0$ for $t \neq 0$ and $\int\delta = 1$. (i) Step from impulse: $u(t) = \int_{-\infty}^{t}\delta(\tau)d\tau$ — for $t < 0$ the integration window does not contain the impulse (0), for $t > 0$ it does (1). (ii) Impulse from step: the smoothed step u_Δ rises from 0 to 1 over $[0, \Delta]$; its derivative is δ_Δ, so $\delta(t) = \lim du_\Delta/dt = du/dt$ (in the generalized sense). Sampling property: since $\delta(t - a) = 0$ except at $t = a$, $x(t)\delta(t - a) = x(a)\delta(t - a)$; integrating, $\int x(t)\delta(t - a)dt = x(a)\int\delta(t - a)dt = x(a)$. (Rigorous version: $\int x(t)\delta_\Delta(t - a)dt = (1/\Delta)\int_a^{a+\Delta} x\,dt \to x(a)$ for continuous x.)

(g) $u(t)u(t - 3) = u(t - 3)$. $u(\cos t) = 1$ whenever $\cos t > 0$, i.e. a square wave of period 2π that is 1 on $(-\pi/2 + 2k\pi, \pi/2 + 2k\pi)$. Its integral from $-\infty$ diverges, so take the integral from 0 (the printed lower limit is not legible; state this assumption):

```
y(t) = ∫₀^t x dτ = u(t-1) + r(t-3) + w(t),   t ≥ 0
w(t) = ∫₀^t u(cos τ)dτ:  t on [0, π/2];  π/2 on [π/2, 3π/2];  π/2 + (t − 3π/2) on [3π/2, 5π/2];
…
       (a staircase-ramp rising with slope 1 for half of each period, average slope 1/2)
y(t/2): replace t by t/2 — every break point doubles (1 → 2, 3 → 6, π/2 → π, 3π/2 → 3π, …)
```

(h) Results:

```
∫(t²+3t+2)δ(−1−t)dt = (t²+3t+2)|_{t=−1} = 1 − 3 + 2 = 0          [δ(−1−t) = δ(t+1)]
∫e^{t−1} sin{π(t−5)/2} δ(t+3)dt = e^{−4} sin(−4π) = 0
((jω+2)/(ω²+9)) δ(ω) = (2/9) δ(ω);       δ(ω−3)/(jω+2) = δ(ω−3)/(2 + j3)
∫_{−5}^{5} δ′(1−τ) cos2πτ dτ = −∫δ′(τ−1)cos2πτ dτ = +d/dτ[cos2πτ]_{τ=1} = −2π sin 2π = 0
∫e^{−5t+1} δ′(t−5) dt = −d/dt[e^{−5t+1}]_{t=5} = 5e^{−24}
Σ_{k=−4}^{4} δ(3t−k) sin²(πt/3) = Σ (1/3) sin²(πk/9) δ(t − k/3): 8 nonzero impulses (k = 0
vanishes).
  Each impulse has finite area but |δ|² is not integrable → E = ∞; with a finite number of
impulses the
  time-average P = lim E/2T is not finite either → neither (state the reason).
```

Lesson 5 — Exponentials, Sinusoids and Periodicity

Slides 30–50 (Lec 2)

1. Real Exponentials and the Step as a Switch (sl. 31–35)

$x(t) = Ce^{at}$: grows ($a > 0$), constant ($a = 0$), decays ($a < 0$). Multiplying by $u(t)$ keeps only the part $t > 0$; $u(t - 2)$ switches on at $t = 2$; $u(3 - t)$ keeps $t < 3$; $u(-t)$ keeps $t < 0$.

```
e^{t}u(-t): rising exponential, ends at 1 at t = 0     e^{-t}u(t-2): starts at e^{-2} at t = 2
e^{t}u(3-t): grows until t = 3, then 0                   e^{-t}u(3t) = e^{-t}u(t)
```

2. Sinusoids and Complex Exponentials (sl. 36–42)

$$x(t) = A\cos(\omega_0 t + \varphi), \quad T_0 = 2\pi/|\omega_0| \qquad e^{j\theta} = \cos\theta + j\sin\theta$$
$$\cos\theta = (e^{j\theta} + e^{-j\theta})/2 \qquad \sin\theta = (e^{j\theta} - e^{-j\theta})/2j$$

$e^{j\omega_0 t}$ is a point rotating around the unit circle; cos and sin are its horizontal and vertical coordinates. A real cosine is two opposite rotations whose imaginary parts cancel (sl. 41). Period: $e^{j\omega_0 T} = 1 \Rightarrow \omega_0 T = 2\pi$.

3. Discrete-Time Periodicity (sl. 43–50)

$x[n] = \cos(\omega_0 n + \varphi)$ is periodic only if some integer N satisfies $\omega_0 N = 2\pi m$. So **$2\pi/\omega_0$ must be rational**: write $2\pi/\omega_0 = N/m$ in lowest terms; the fundamental period is N (after m cycles of the underlying cosine).

Signal	$2\pi/\omega_0$	Result
$\cos(2t)$ (CT)	—	periodic, $T_0 = \pi$
$\cos(2n)$	π (irrational)	not periodic
$\cos(2\pi n/5)$	5	$N = 5$
$3\sin(5n/7)$	$14\pi/5$ (irrational)	not periodic
$2\cos(16\pi n/63 + \pi/3)$	63/8	$N = 63$ (8 cycles of the cosine)

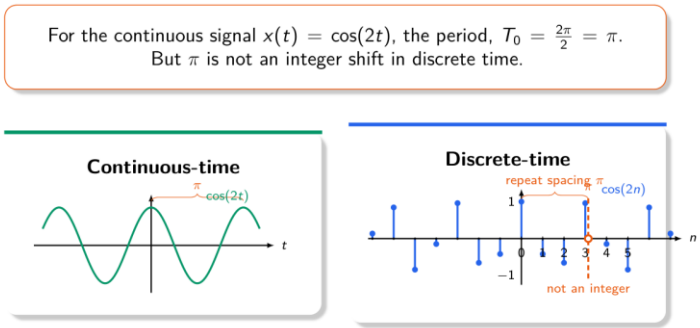


Figure 5.1 — cos(2t) repeats every π, but cos(2n) never repeats: π is not an integer shift (Lec 2, sl. 45)

4. Sums of Periodic Signals (textbook)

$x_1 + x_2$ (periods T_1, T_2) is periodic iff T_1/T_2 is rational; then $T = \text{LCM}(T_1, T_2)$. Equivalently ω_1/ω_2 rational and $\omega_0 = \text{GCD}(\omega_1, \omega_2)$. For DT sums: $N = \text{LCM}(N_1, N_2)$ always exists if each part is periodic. Phase shifts never change a period. Powers and products: expand into sums of sinusoids first ($\sin^2\theta = (1 - \cos 2\theta)/2$).

TRAP • The phase of a DT sinusoid does not affect periodicity (sl. 49). • In DT, $\omega_0 = 2\pi\cdot(m/N)$ gives period N, not $2\pi/\omega_0$ (63/8 is not a period). • $\sqrt{(e^{j5\pi t})} = e^{j5\pi t/2}$: halve the frequency. • An impulse train $\delta(t - k\pi)$ has period π, $u(\cos t)$ has period 2π.

Exam Approach

1. Rewrite everything as a sum of sinusoids / complex exponentials (expand squares and products).
2. CT: list the periods $T_i = 2\pi/\omega_i$; check every ratio is rational; T = LCM.
3. DT: for each term check $2\pi/\omega_i = N/m$ rational; N_i = numerator; N = LCM(N_i).
4. Write the conclusion with the reason ("ratio is irrational ⇒ not periodic").

Practice — Lesson 5

(a) Actual — CSE 219 2022-23 Q5(c). (b) Actual — CSE 219 2023-24 Q3(a). (c) Actual — EEE 211 2021-22 Q1(a). (d) Older papers, quick answers (EEE 211 2017-18 Q5(a), 2016-17 Q5(c); EEE 301 2015-16 Q1(a), 2014-15 Q1(a), 2013-14 Q1(a), 2012-13 Q1(a), 2011-12 Q6(a), 2010-11 Q5(a)(i)).

(a) Is $x[n] = 2\cos(16\pi n/63 + \pi/3)$ periodic? Explain. (5)

(b) $x[n] = (3 - 4j)e^{j13\pi n/7}$. (i) Are the real and imaginary parts periodic? Fundamental period? (ii) Can the phase shift in each part be interpreted as an integer time shift of a cosine sequence? (15)

(c) Is $x(t) = \sqrt{(e^{j5\pi t})} + \sin^2(3t - \pi/4)$ periodic? If so, find the period. (10)

(d) Periodic? Period? (i) $[2\cos(0.6\sqrt{\pi}\, t + \pi/6) + \sin(1.2\sqrt{\pi}\, t - \pi/3)]^2$; (ii) $\cos(100\pi t + \pi/3) + \sin(40\pi t + 2\pi/3)$; (iii) $3\cos(4t + \pi/3)$, $[\cos(2t - \pi/3)]^2$, $\Sigma_{n=-7}^{7} e^{-(2t-n)}$, $e^{j(\pi t-1)}$; (iv) $u(\cos t) - \delta(t - k\pi)$; (v) $4\cos \pi t - \sin(3\pi t + \pi/2)$ and $4\cos 2t + 4\cos 2\pi t$; (vi) $\alpha\delta(\sin t) + (1 - \alpha)\delta(t - k\pi)$, k odd, $0 < \alpha < 1$; (vii) $e^{j7\pi t/6} + e^{j5\pi t/6}$; (viii) $\cos(\pi t/3) + \sin(\pi t/4)$.

✓ Solution

(a) $\omega_0 = 16\pi/63$, $2\pi/\omega_0 = 63/8 = N/m$ with N = 63, m = 8 (rational). **Periodic, $N_0 = 63$**: after 63 samples the angle advances by $16\pi = 8$ full turns. The phase $\pi/3$ does not matter.

(b) $3 - 4j = 5e^{-j\theta}$, $\theta = \tan^{-1}(4/3) = 0.9273$ rad. $x[n] = 5e^{j(13\pi n/7 - \theta)}$, so Re x = $5\cos(13\pi n/7 - \theta)$, Im x = $5\sin(13\pi n/7 - \theta)$. (i) $2\pi/\omega_0 = 2\pi/(13\pi/7) = 14/13 \rightarrow$ rational, N = 14 (13 cycles). Both parts are periodic with **$N_0 = 14$**. (ii) A time shift n_0 turns $\cos(13\pi n/7)$ into $\cos(13\pi(n - n_0)/7)$, a phase of $-13\pi n_0/7$, which is always a multiple of $\pi/7$ (mod 2π). $\theta = 0.9273$ rad = $2.066\cdot(\pi/7)$ is not a multiple of $\pi/7$ (in fact θ/π is irrational, since $\cos\theta = 3/5$ is rational but not 0, ±½, ±1). So **no**: the phase cannot be produced by an integer shift — it can only be read as a shift of the underlying CT cosine.

(c) $\sqrt{(e^{j5\pi t})} = e^{j5\pi t/2}$ (principal root): period $2\pi/(5\pi/2) = 4/5$. $\sin^2(3t - \pi/4) = ½ - ½\cos(6t - \pi/2)$: period $\pi/3$. Ratio $(4/5)/(\pi/3) = 12/(5\pi)$ is irrational ⇒ **not periodic**.

```
(d)(i)   periods 10√π/3 and 5√π/3 (ratio 2); the square contains 0.6√π·{1,2,3,4} ⇒ T = 10√π/3
≈ 5.91
(ii)     T₁ = 1/50, T₂ = 1/20 ⇒ T = LCM = 0.1 s
(iii)    π/2;  π/2 (cos² = ½ + ½cos(4t − 2π/3));  e^{−2t}·const → not periodic;  T = 2
(iv)     u(cos t): 2π;  impulse train: π  ⇒ T = 2π
(v)      T = 2 and 2/3 ⇒ T = 2;    π and 1 (ratio π irrational) ⇒ not periodic
(vi)     δ(sin t) = Σ_k δ(t − kπ) (|cos kπ| = 1) ⇒ weight α at even multiples of π, α + (1−α) =
1 at odd
         multiples ⇒ periodic, T = 2π (α ≠ 1). For α = 0.5: ½ at 0, ±2π…; 1 at ±π, ±3π…
(vii)    ω₀ = GCD(7π/6, 5π/6) = π/6 ⇒ T = 12
(viii)   T₁ = 6, T₂ = 8 ⇒ T = 24
```

Quick Notes — Signals

Revision points for Lessons 1–5.

L1 · Energy and Power

- $E = \int|x|^2dt$ ($\Sigma|x[n]|^2$); $P = E$ / interval length; $P_\infty = \lim (1/2T)\int_{-T}^{T}|x|^2$.
- Energy signal: E finite, P = 0. Power signal: P finite, E = ∞. Neither: both infinite.
- Periodic → power, average one period. $A\cos(\omega_0 t + \varphi)$: $P = A^2/2$; $Ae^{j\omega_0 t}$: $P = |A|^2$.
- Half-sine E = π/2; 1 (|t| ≤ 1), 1/|t| beyond: E = 4.

L2 · Transformations

- x(t − t0) delay (right); x(t + t0) advance (left); x(−t) mirror; x(αt) compress (α > 1).
- Recipe for x(αt + β): shift by β → reverse if α < 0 → scale by |α|. Name each intermediate signal.
- Feature at a moves to t = (a − β)/α. x(2t + 3) ≠ x(2(t + 3)).
- 2022-23 Q7: x(−2t+2) = t [0,1], 1 [1,1.5], 3 − 2t (1.5,2]; x(−t/2+1) = ½ + t/4 [−2,2], 1 [2,4], 2 − t/2 (4,6].

L3 · Even and Odd

- $x_e = \frac{1}{2}[x(t) + x(-t)]$, $x_o = \frac{1}{2}[x(t) - x(-t)]$; work interval by interval.
- x = t+1 on [−1,1] → $x_e = 1$, $x_o = t$. u[n] → $x_e = \frac{1}{2}$ (1 at 0), $x_o = \pm\frac{1}{2}$.
- 2022-23 Q5(a): $f_e = 1 - 3|t|/4$, $f_o = t/4$ on |t| ≤ 2.

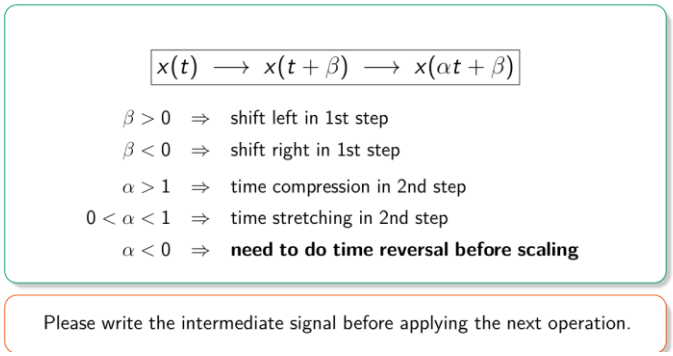
L4 · Step, Impulse, Ramp

- $\delta[n] = u[n] - u[n-1]$; $u[n] = \Sigma_{m\le n}\,\delta[m] = \Sigma_{k\ge 0}\,\delta[n-k]$. $\delta(t) = du/dt$; $u(t) = \int_{-\infty}^{t}\delta$.
- Jump of J at a → $J\delta(t-a)$ in the derivative; slope change m → $m\cdot r(t-a)$ in the signal.
- Sifting $\int x\,\delta(t-a) = x(a)$ only if a is inside the limits; $\delta(at) = \delta(t)/|a|$; $\int x\,\delta'(t-a) = -x'(a)$.
- δ model: unit area for all ε, → 0 for t ≠ 0, peak → ∞. $\sin(\varepsilon t)/(\pi t)$ fails as ε → 0.

L5 · Periodicity

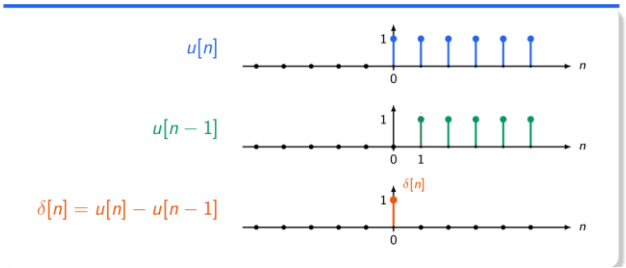
- CT: $T_0 = 2\pi/|\omega_0|$; sum periodic iff period ratio rational, T = LCM.
- DT: periodic iff $2\pi/\omega_0 = N/m$ rational; N = numerator. cos 2n not periodic; $2\cos(16\pi n/63 + \pi/3)$: N = 63.
- Phase never changes a period. Expand squares/products before deciding.

Key Diagrams to Reproduce



The x(αt + β) recipe (Lec 1, sl. 30)

$$\delta[n] = u[n] - u[n-1]$$



δ[n] = u[n] − u[n − 1] (Lec 2, sl. 11)